Awarding Body:	ARDEN UNIVERSITY
Programme Name:	BSc in Computing with Foundation Year
Module Name (and Part if applicable):	System Analysis and Design
Assessment Title:	System Analysis & Design Case Study Report
Student Number:	STU142158
Tutor Name:	Arif Mushtaq
Word Count:	3971

Please refer to the Word Count Policy on your Module Page for guidance

Contents

Table of Figures

Introduction

The current operational challenges of Prime Estates Ltd. are due to outdated and fragmented property management systems. To this end, the company intends to develop a centralized Property Management System (PMS) to increase efficiency and uniformity across all the branches. However, to ensure success in the implementation of the change, choosing the right Software Development Life Cycle (SDLC) model is important to match with the company's business needs, constraints and technical capabilities.

The Agile SDLC model is recommended for this project, and this is illustrated by how the model meets Prime Estates' requirements, involves stakeholders, and ensures rapid deployment while meeting business needs. This choice is supported by a comparative analysis with the Waterfall model, as well as flexibility, cost-effectiveness, and risk mitigation strategies.

Key Challenges & Business Constraints

As Prime Estates Ltd. moves to implement the new PMS, several operational challenges in the adoption process need addressing:

1. **Fragmented and Outdated Systems:** The use of local Microsoft Access databases along with manual processes produces inconsistent data while restricting business scalability.

2. **Limited Technical Expertise:** Estate agents especially older staff members show limited understanding of new technologies thus they need essential training for effective system adoption (Kendall & Kendall, 2019)

3. **Resistance to Change:** Staff members exhibit resistance because they fear both complex systems and disruptions thus a planned transition approach is necessary (Valacich et al., 2015).

4. **Time Constraints:** The business owner demands a six-month deployment for the full system with an initial functional version to maintain business operations throughout (Improving, 2023).

5. **Budget Limitations**: The company needs to guarantee cost-effective implementation which prevents future system upgrades while ensuring both quality and scalability (Sommerville, 2016).

6. **Legal & Accessibility Compliance:** The system must meet data protection standards (e.g., GDPR) and include features for employee accessibility for people with disabilities according to Cadle (2014).

7. **Integration with External Platforms:** The PMS needs full integration with National property websites so that property listings get automated while minimizing manual errors according to Agile Alliance (2022).

Selection of SDLC Model: Agile vs. Waterfall

The following are some operational challenges that Prime Estates Ltd. is facing due to outdated and fragmented property management processes. As a solution, a centralized Property Management System (PMS) is recommended. This is because selecting the right Software Development Life Cycle (SDLC) model is very important to achieve a smooth transition, stakeholder involvement, and successful deployment of the system within the given resources.

Recommended SDLC Model: Agile Approach

Prime Estates Ltd should implement the Agile SDLC model because it is flexible, iterative, has stakeholder involvement, and produces a system that is aligned with user requirements (Agile Alliance, 2022). The agility of Agile helps in better adaptation to the changing business needs for better efficiency. The following table shows how Agile addresses the critical challenges:

Challenge	How Agile Solves It
Fragmented & Outdated Systems	Early **MVP deployment** minimizes disruption, with **incremental improvements** ensuring seamless transition (Sommerville, 2016).
Limited Technical Knowledge	Regular **feedback sessions** ensure a **user-friendly design**, reducing resistance (Kendall & Kendall, 2019).
Resistance to Change	**Gradual implementation through sprints** allows employees to adapt, reducing fear of disruption (Valacich et al., 2015).
Time Constraints (6 months)	Sprint-based model delivers **functional components in weeks**, keeping the project on track (Improving, 2023).
Budget Limitations	Prioritizes **critical features in early iterations**, allowing **cost adjustments** based on business needs (Sommerville, 2016).
Legal & Accessibility Compliance	**Continuous testing cycles** ensure GDPR compliance and accessibility (e.g., screen reader support) (Cadle, 2014).
Integration with External Platforms	**Modular API integrations** automate property postings, reducing errors (Agile Alliance, 2022).

Figure 1:Agile SDLC Solutions for Prime Estates Ltd.'s Challenges.

The agility of the approach ensures that Prime Estates Ltd. can create an efficient, scalable, and user-friendly Property Management System, compliant with the need to optimize.

Comparison with the Waterfall Model

The Waterfall model is too inflexible for Prime Estates Ltd. and the frequent updates in property management processes require (Sommerville, 2016). Because Waterfall requires fully defined requirements at the beginning, any change in the process during development, for instance, incorporating mobile access for estate agents, would be costly and time-consuming (Kendall & Kendall, 2019). Furthermore, Waterfall does not have early testing and user involvement which is a risk in terms of usability particularly given the limited technical knowledge of the staff (Valacich et al., 2015). It would only manifest at the final stage, which would mean costly redesigns and retraining. On the other hand, Agile is integrated with real-time stakeholder feedback, which allows the system to develop according to actual business needs (2022 Agile Alliance).

Agile is therefore suitable for the Prime Estates Ltd. in the development of the new Property Management System because it allows for flexibility, integration of features and stakeholder participation at the different stages of development.

Feature	Agile SDLC (Recommended)	Waterfall SDLC
Flexibility	Adapts to evolving needs **mid-development** (Kendall & Kendall, 2019).	**Rigid structure**; changes require restarting development.
Stakeholder Engagement	Continuous feedback improves usability and **reduces resistance** (Valacich et al., 2015).	Stakeholders are involved only **at the start and end**, risking misalignment.
Deployment Speed	**MVP delivered quickly** with **continuous updates** (Improving, 2023).	Full system is delivered only **after all phases**, delaying business benefits.
Risk Mitigation	Ongoing testing prevents **major failures** (Sommerville, 2016).	Late-stage testing increases risk of **costly errors**.
Scalability	Supports **future upgrades & modular features** (Agile Alliance, 2022).	Difficult to modify post-development without **significant rework**.

Figure 2:Comparison of Agile and Waterfall SDLC Models

As such, these factors favour Agile, which is Prime Estates' operational, resource constraints, and business fit.

Potential Risks of Agile and Mitigation Strategies

Agile provides many benefits but comes with its own collection of challenges. The following table shows potential risks and their corresponding mitigation strategies to help guarantee a smooth implementation.

Risk Factor	Potential Impact	Mitigation Strategy
Scope Creep (Frequent addition of new features)	Increased costs, delayed system completion.	Define a **strict MVP scope**; introduce **change request approval processes**.
Limited Stakeholder Availability (Estate agents too busy for frequent feedback)	Inconsistent feedback may lead to **missed requirements**.	Appoint **branch representatives** for consolidated feedback; schedule fixed **bi-weekly review meetings**.
Documentation Challenges (Focus on working software over documentation)	Difficulties in system maintenance and onboarding new employees.	Maintain **lightweight but structured documentation**, including key workflows and API integrations.
Adoption Resistance from Low-Tech Users	Employees may struggle with system usability.	Provide **step-by-step tutorials**, **hands-on training sessions**, and 24/7 user support.

Figure 3:Challenges of Agile and Mitigation Strategies

Through the implementation of these mitigation strategies, the risks of Agile can be minimized, and the benefits can be maximized for optimal management.

Case Studies Supporting Agile Adoption

Real-world implementations of Agile further reinforce its effectiveness:

NASA's TReK Project – Agile was used to increase flexibility and reduce risk, and in the process was successful in complex software systems (Hendrix & Schneider, 2002).

TSB Bank IT Failure – This is an example of a poorly planned and implemented rollout of a system using the waterfall model, which resulted in a catastrophic system failure that affected millions of customers (Peachy, 2018). This highlights Agile's advantage in reducing deployment risks.

UK Government Digital Transformation – This has used Agile to modernise public service platforms, with efficient rollouts and user focused enhancements (GOV.UK, 2022).

Prime Estates Ltd. should use Agile as the SDLC model of choice because it provides for the rapid delivery of working software, takes in stakeholder feedback, and deals with operational inefficiencies. Even such risks as scope creep and adoption challenges as possible risks can be overcome using structured backlog management, phased deployment, and targeted training programs.

The application of Agile will preserve the competitive edge of Prime Estates Ltd., improve its operational efficiency, and address the changing expectations of its clients.

Requirements Analysis for the Property Management System (PMS)

For the company Prime Estates Ltd., the implementation of a centralized Property Management System (PMS) has been recommended to address operational inefficiencies. In this part of the research, is present a systematic approach for the analysis of the system that can fulfil business requirements, technical capabilities, and legal norms (Kendall & Kendall, 2019).

In addition, the approach follows practitioner-level design techniques, ensuring that:

- Business workflows can be enhanced because of well-defined system functionalities.
- The actors and use cases have been correctly related to user roles for alignment.
- Use case diagrams and detailed workflows give a clear idea of system behavior.
- Compliance and security measures are integrated into the design.
- Optimization strategies have been adopted to guarantee scalability and efficiency.

This analysis ensures the PMS is effective, scalable, and relevant to real-world business challenges by applying industry-standard methodologies (Sommerville, 2016).

Core Functionalities of the PMS

The PMS must deliver on Prime Estates Ltd.'s main operational objectives through defined core functionalities. A structured functional approach enhances usability, reduces errors, and automates repetitive tasks (Fox, 2017). The five core functionalities were chosen for their ability to meet the most important business requirements while addressing client management through to compliance.

1. Generating Clients and Property

(Attracting and managing potential buyers, tenants, and landlords)

Includes:

- **Property Listings & Advertisements Automated** (Makes certain properties are listed properly and extensively.)
- **CRM** (Customer Relationship Management) **for Clients & Landlords** (Record client interactions and follow up on leads)
- **Performance Analytics on Listings** (The properties that attract most interest are pinpointed.)
- **Mobile Access for Agents** (Agents can update listings from their mobile devices).

2. Handling Administration

(Controlling internal processes, contracts, and operational effectiveness)

Includes:

- **Centralized Property & Client Database** (Ensures that all information is well documented and easily accessible at the right time and place)
- **Automated Brochure Generation** (Reduces the time and labour spent on creating marketing materials by the company)
- **Document & Contract Storage** (Makes it easier to create, manage and
- retrieve agreements.)
- **Financial Management** (Invoice Generation & Payment Tracking) (Prevents errors in payments and invoices that can lead to legal issues).
- **User Role Management & Security** (Makes sure that sensitive information is only accessible to the right people)
- **Legal Compliance & Data Protection** (GDPR, Accessibility) (Ensures compliance with laws and regulations such as GDPR and accessibility standards.

3. Managing Sales

(Handling property transactions, negotiations, and closing deals)

Includes:

- **Multi-Branch Landlord Management** (Tracks landlords with multiple properties)
- **CRM** (Customer Relationship Management) **for Sales Tracking** (Stores client inquiries, offers, and deal progress)
- **Property & Pricing Management** (Ensures updated property details for buyers)
- **Automated Contract Generation & Digital Signing** (Speeds up agreement processes)

4. Managing Rentals

(Handling lease agreements, tenants, and property maintenance)

Includes:

- **Tenant & Lease Agreement Management** (Tracks lease renewals and tenant information)
- **Automated Rental Payment Tracking** (Reminders for on time payments)
- **Online Maintenance Request System** (Tenants can place and follow up on requests)
- **Landlord & Tenant Communication Features** (Document all rental related discussions).

5. System Integration

(Ensuring seamless connectivity with external systems and platforms)

Includes:

- **National Property Websites:** Properties are directly linked to Rightmove and Zoopla, information such as listings, availability, and status of sold or rented properties are updated in real time.
- **Advertisement Fee Tracking** – This links advertisement payments to financial records and notifies financial staff of overdue payments.
- **Error Handling & Notifications** – This application will also monitor for synchronization errors and notify the user of the same for the necessary action to be taken.
- **Performance Metrics from External Platforms** – This helps in monitoring the performance of the listings for example the number of views and inquiries on the integrated property websites which in turn helps the estate agents to enhance their marketing.

Actors and Use Cases

The real-world user roles should be aligned with system design through clear definition of actors which enables more effective role-based access control so that each user only interacts with relevant system features (Sommerville, 2016).

Actor	Role & Responsibilities
Estate Agent	Manages property listings, client interactions, sales, and rentals.
Financial Staff (Head Office)	Manages invoices, advertising payments, and financial transactions.
IT Technician (Head Office)	Provides system support, maintenance, and security.
Tenant (Property Renter/Buyer)	Submits inquiries, books viewings, and reports maintenance issues.
Landlord (Property Owner)	Provides property details, tracks rental payments, and signs agreements.
Contractors (Maintenance & Repair Services)	Receives maintenance requests and updates repair status.

Figure 4: *PMS Actors and Their Roles*

Use Case Diagram

The Use Case Diagram (Figure 4) presents system functionalities together with actor interactions that produce well-defined workflow.

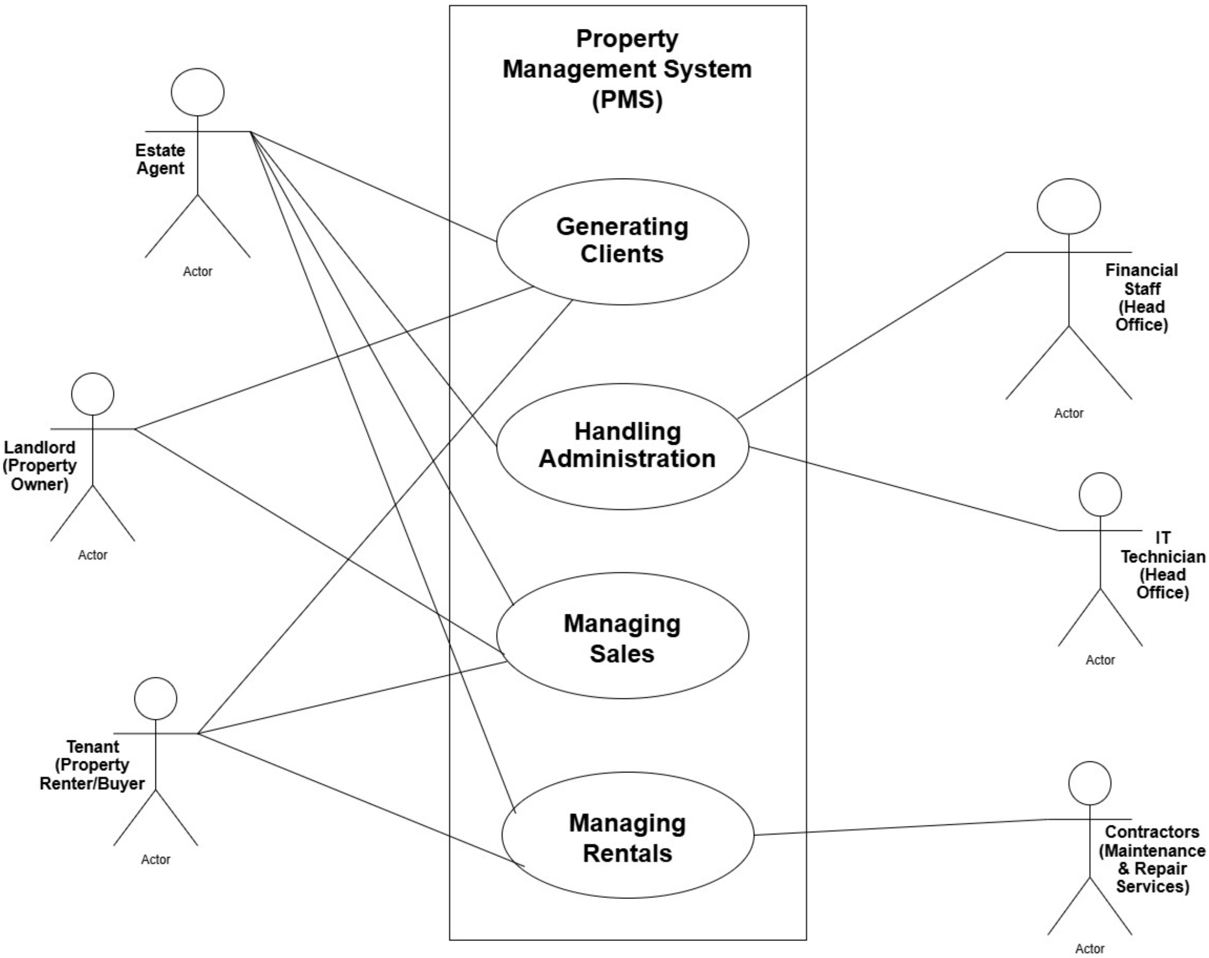


Figure 5:PMS Use Case Diagram

Use Case Course of Events Tables

To guarantee accurate representation of business processes within the system while meeting stakeholder requirements through defined system actions and maintaining robustness through error handling and alternate flows – each use case scenario was documented in detail. The following tables (Figures 5–8) detail the steps for four core functionalities:

Use Case 1: Generating Clients and Property Listings

Use Case: Generating Clients and Property **ID:** 1
Importance Level: High
Use Case Type: Detail, Essential

Primary Actor: Estate Agent

Other Actors: Landlord, Tenant, Buyer

Stakeholders and Interests:

- **Estate Agents** – Want to efficiently attract and onboard clients, ensuring accurate profiles for targeted follow-ups.
- **Landlords** – Want their properties listed widely and correctly to attract tenants or buyers.
- **Tenants** – Want to register interest in properties and receive updates.
- **Buyers** – Want tailored property recommendations and seamless communication.

Brief Description: This use case describes how estate agents attract and onboard new clients (landlords, tenants, or buyers) and property, ensuring proper record-keeping and targeted communication.

Trigger: A landlord, tenant, or buyer submits an inquiry through the portal, phone, or in-person.

Type: External

Relationships:

- **Association:** Centralized Property & Client Database
- **Include:** Validation of client details
- **Extend:** Automated communication setup (e.g., email updates)
- **Generalization:** None

Normal Flow of Events:

1. A landlord, tenant, or buyer submits their details via the portal or in person.
2. The estate agent logs into the PMS and selects the type of client (landlord, tenant, or buyer).
3. The agent enters the client's and properties information into the system.
4. The system validates the data and stores it in the **Property & Client Database**.
5. The system synchronizes property details with the **National Property Websites** (System Integration).
6. The system sends a confirmation email or SMS tailored to the client type.
7. The client's profile is added to the CRM for performance tracking and follow-up communication.

Alternate/Exceptional Flows:

- **If client details are incomplete:**
  - The system prompts the agent to provide missing information.
- **If synchronization with external websites fails:**
  - The system retries the process and notifies the agent.

Figure 6: Generating Clients and Property Listings Use Case

Use Case 2: Handling Administration & Compliance

Use Case: Handling Administration **ID:** 2
Importance Level: High
Use Case Type: Detail, Essential

Primary Actor: Financial Staff

Other Actors: Estate Agent, IT Technician

Stakeholders and Interests:

- **Financial Staff –** Want automated tools for invoicing, payment tracking, and managing financial operations.
- **Estate Agents** – Need quick access to contracts and client documents.
- **IT Technicians** – Ensure system security, compliance, and functionality.

Brief Description: This use case focuses on managing financial tasks, contracts, and system compliance, streamlining internal administrative processes.

Trigger: A financial staff member or estate agent logs into the system to manage records, payments, or documents.

Type: Internal

Relationships:

- **Association:** Centralized Property & Client Database
- **Include:** Automated invoicing and GDPR compliance checks
- **Extend:** Role-based access and audit logging
- **Generalization:** None

Normal Flow of Events:

1. Financial staff log into the PMS and access the administration module.
2. The system displays pending invoices, payments, and tasks requiring attention.
3. Staff generate invoices for landlords or tenants, using automation tools.
4. Estate agents retrieve contracts or documents from the **Centralized Document Storage**.
5. The system ensures data security and role-based access compliance.
6. IT technicians monitor user activities and enforce GDPR compliance.

Alternate/Exceptional Flows:

- **If invoice data is incomplete:**
  - The system flags it for review and notifies financial staff.
- **If unauthorized access is detected:**
  - The system blocks access and notifies IT technicians.

Figure 7: Handling Administration & Compliance Use Case

Use Case 3: Managing Sales

Use Case: Managing Sales **ID:** 3
Importance Level: High
Use Case Type: Detail, Essential

Primary Actor: Estate Agent

Other Actors: Landlord, Buyer

Stakeholders and Interests:

- **Estate Agents** – Want streamlined sales processes for tracking offers and closing deals.
- **Landlords** – Want up-to-date information about sales progress.
- **Buyers –** Expect a smooth and efficient property purchasing process.

Brief Description: This use case manages property sales, from tracking inquiries to closing deals, ensuring efficiency and transparency for both buyers and landlords.

Trigger: A buyer submits an inquiry or makes an offer.

Type: External

Relationships:

- **Association:** Client Database, Property Database
- **Include:** CRM for tracking inquiries and offers
- **Extend:** Automated contract generation
- **Generalization:** None

Normal Flow of Events:

1. A buyer submits an inquiry via the portal or in person.
2. The estate agent retrieves property and buyer details from the PMS.
3. The agent updates the CRM with the inquiry and any subsequent offers.
4. The estate agent negotiates with the buyer and updates the PMS with offer details.
5. The system validates offer details and generates a sales contract automatically.
6. The landlord reviews and signs the contract digitally.
7. The buyer completes the purchase, and the system updates the property status to "Sold."

Alternate/Exceptional Flows:

- **If a buyer retracts their offer:**
  - The CRM flags the inquiry as "inactive."
- **If multiple offers are made:**
  - The system tracks all offer for the agent to negotiate with buyers.

Figure 8:Managing Sales Use Case

Use Case 4: Managing Rentals

Use Case: Managing Rentals **ID:** 4
Importance Level: High
Use Case Type: Detail, Essential

Primary Actor: Estate Agent

Other Actors: Tenant, Landlord, Contractors

Stakeholders and Interests:

- **Estate Agents** – Need to manage tenant inquiries, lease agreements, and property maintenance efficiently.
- **Landlords** – Expect timely rental payments and maintenance issue resolution.
- **Tenants** – Want a seamless rental process and easy access to property services.
- **Contractors** – Want clear and timely maintenance task assignments.

Brief Description: This use case involves managing the rental lifecycle, from tenant inquiries to maintenance requests, ensuring smooth and efficient operations.

Trigger: A tenant submits an inquiry or maintenance request.

Type: External

Relationships:

- **Association:** Tenant Database, Maintenance Database
- **Include:** Rental payment tracking
- **Extend:** Maintenance request management
- **Generalization:** None

Normal Flow of Events:

1. A tenant submits an inquiry about renting a property.
2. The estate agent matches the tenant with suitable properties and schedules a viewing.
3. The tenant submits a rental application, which the system validates.
4. The system generates a rental agreement for both tenant and landlord to sign digitally.
5. The tenant begins making rental payments, tracked by the system.
6. The tenant submits maintenance requests via the portal when required.
7. The system assigns maintenance tasks to contractors and tracks completion.

Alternate/Exceptional Flows:

- **If a tenant's application is incomplete:**
    - The system prompts the tenant to provide missing details.
- **If maintenance is delayed:**
    - The system escalates the issue to the estate agent.

Figure 9: managing Rentals Use Case

Non-Functional Requirements

Non-functional requirements are how the system performs, including security, scalability, and legal compliance (National Cyber Security Centre, 2019).

Category	Requirement
Performance	The system shall handle up to **1,000 concurrent users** without performance degradation (*Sommerville, 2016*). Response time for key operations (e.g., generating reports, updating client records) shall not exceed **2 seconds** under normal load (*Sommerville, 2016*).
Scalability	The system shall be **scalable to accommodate future growth**, including additional branches and increased property listings. It shall support **cloud-based deployment** to enable seamless scaling of resources.
Usability	The system shall have an **intuitive user interface** to accommodate users with **limited technical expertise** (*Kendall & Kendall, 2019*). It shall include **onboarding tutorials and contextual help** to assist new users (*Kendall & Kendall, 2019*).
Reliability	The system shall have an uptime of **99.9%**, ensuring minimal disruption to operations. It shall include **automated backup and recovery mechanisms** to prevent data loss.
Security	The system shall implement **Multi-Factor Authentication (MFA)** to enhance user account security (*Microsoft, 2022*). It shall **encrypt sensitive data** (e.g., client details, financial records) both **in transit and at rest** (*National Cyber Security Centre, 2019*).
Accessibility	The system shall comply with **WCAG 2.1 standards** to ensure accessibility for users with disabilities, such as vision impairments (*National Cyber Security Centre, 2019*). It shall include features like **screen reader compatibility and keyboard navigation** (*National Cyber Security Centre, 2019*).

Figure 10: Non-Functional Requirements of the PMS

Compliance & Regulatory Factors

Companies must adhere to legal and cybersecurity standards to protect sensitive customer data and prevent legal risks according to National Cyber Security Centre (2019).

Regulation	Requirement
GDPR (EU Law)	Ensures **secure data handling and user privacy** (GOV.UK, 2018).
National Cyber Security Standards	Requires **data encryption and security monitoring**.

Figure 11: PMS Regulatory Compliance Considerations

Impact of Constraints on System Requirements

The system requirements are determined by the budget and current technology infrastructure of Prime Estates Ltd. Since the owner does not want to replace the computers and wants a cost-effective solution, the PMS should be compatible with the current hardware and should be a lightweight web-based system instead of a resource-intensive one (Sommerville, 2016).

The budget also has a effect on the feature prioritization. To control costs, the deployment of advanced tools like AI-driven analytics may be delayed to later phases (Valacich et al., 2015). Moreover, because members of staff have low levels of technical knowledge, the system should have a simple interface to keep training costs low and increase adoption (Cadle, 2014).

The constraints of budget and staff expertise are addressed by agile development which enables phased implementation that aligns features with budget availability and provides for scalability (Agile Alliance, 2022).

The core functional requirements are the essential features of the PMS, whereas the non-functional requirements include reliability, security, and efficiency of the system. Suggesting the following optimizations, Prime Estates Ltd. can get the most out of the system, improve business processes, and improve customer satisfaction. (Sommerville, 2016)

Optimization Strategies for the Property Management System (PMS)

For the Property Management System (PMS) of Prime Estates Ltd. to function efficiently, grow with the company and provide long-term benefits, it is crucial to implement certain key optimization measures. The strategies focus on automation, AI-driven analytics, centralized data management, integration, and performance monitoring to make the PMS robust, cost-efficient, and user-friendly.

Automation of Manual Tasks

Workflow management becomes more efficient through automation while human error reduction and process optimization contribute further to these efficiency gains. The elimination of repetitive manual processes by Prime Estates Ltd. leads to better overall productivity and reduced processing delays.

Automation of Manual Tasks

Task	Implementation Plan
Automated Invoice Generation & Payment Tracking	Implement rule-based workflows, integrate with PayPal/Stripe, introduce error handling (Kendall & Kendall, 2019).
Automated Property Listings	Develop API for Rightmove/Zoopla, enable real-time updates, data validation (Agile Alliance, 2022).
Automated Maintenance Requests	Self-service portal for tenants, AI-driven issue prioritization, contractor assignment system (Valacich et al., 2015).

AI-Driven Analytics for Business Insights

AI Feature	Implementation Plan
Predictive Analytics for Pricing	Use machine learning models for property value prediction (Sommerville, 2016).
Client Behavior Tracking	Integrate CRM, AI-based lead generation (Cadle, 2014).
Agent Performance Analytics	Real-time performance dashboards, AI-driven training recommendations (Fox, 2017).

Figure 12:Automation of Manual Tasks in PMS Implementation &AI-Driven Analytics for Business Insights

Prime Estates Ltd. will be able to make more accurate and informed decisions on property pricing, client interaction, and agent performance through the integration of Artificial Intelligence (AI) and machine learning.

Centralized Data Management & Real-Time Synchronization

Feature	Implementation Plan
Cloud-Based Database	Migration to PostgreSQL/AWS, ETL processes for data validation (National Cyber Security Centre, 2019).
Real-Time Data Synchronization	Use Apache Kafka/RabbitMQ, enable offline access (GOV.UK, 2018).
Role-Based Access Control	Tiered access, MFA, encryption for security compliance (GOV.UK, 2018).

Seamless Integration with External Platforms

Integration	Implementation Plan
Property Listing Websites	RESTful API for bidirectional synchronization (Kazi & Kazi, 2022).
Accounting & Financial Software	Integration with Xero/QuickBooks, automated transaction reconciliation.
Automated Notification System	SMS/email alerts, AI-driven chatbot for tenant queries (Fox, 2017).

Figure 13: Centralized Data Management & Real-Time Synchronization & Seamless Integration with External Platforms

The PMS must also be able to effectively interact with third-party services such as property listing sites, accounting applications, and payment processors to reduce the likelihood of human mistake and decrease workload.

Performance Monitoring & System Optimization

To guarantee that the PMS is always in top form, secure and scalable, it is crucial that there is continuous system monitoring and optimization.

Monitoring & Optimization	Implementation Plan
System Health Monitoring	Use AWS CloudWatch/New Relic, real-time failure alerts (Sommerville, 2016).
Predictive Maintenance	AI-driven detection for IT infrastructure issues.
User Activity Logging	Audit trails for security compliance, behavior analytics for UI improvement (Valacich et al., 2015).

Figure 14:Performance Monitoring & System Optimization

The implementation of these optimization strategies will enable Prime Estates Ltd. to streamline operations, improve data accuracy and enhance decision-making capabilities. The integration of automation, AI-driven analytics, centralized data management, the seamless integration of multiple platforms and proactive system monitoring will guarantee that the PMS is efficient, secure, and adaptable to the company's future business growth.

System Design: Data Flow Diagrams (DFDs) for the Property Management System (PMS)

In systems analysis and design, Data Flow Diagram (DFD) is a vital tool for visually representing how data flows through a system (Kendall & Kendall, 2019). DFDs illustrate:

- Data inputs and outputs for various system processes.
- Information transformation and storage within different components.
- User-system interactions.

The Data Flow Diagrams (DFDs) show the flow of data within the Property Management System (PMS) for better management of listings, sales, rentals, and administration. However, there is further detail on how particular processes work within the course of the argument. For instance, in the Generating Clients and Property Listings process, the system collects information from clients, checks the information about the properties and updates the listings on external sites such as Rightmove and Zoopla. The Managing Rentals process is for lease agreements and includes tracking the tenant's details, creating the rent invoices, and sending automated reminders (Kendall & Kendall, 2019). These detailed process explanations help to link system functionality to business operations.

Alternative Approaches: Entity-Relationship Diagrams (ERDs)

Although the Data Flow Diagrams (DFDs) are good for representing the data flow, the Entity-Relationship Diagrams (ERDs) offer an alternative perspective by concentrating on the structure of the database and the relationships among important entities such as Landlords, Tenants, Properties, and Payments (Sommerville, 2016). This would help to complement DFDs in ensuring that data storage is in line with the functional workflows of the system.

Using both DFDs and ERDs in system design would give a more detailed picture, so that both process perspectives and structure of the database are well organized to meet the operational requirements of Prime Estates Ltd.

For Prime Estates Ltd.'s Property Management System (PMS), the DFD approach is applied to decompose extensive system workflows into structured, hierarchical layers:

- **Level 0 (Context Diagram):** A high-level overview of the system as one process communicating with external users.
- **Level 1:** This part of the system presents core operations and major subsystems in functional detail.
- **Level 2:** Key processes are expanded and detailed to show workflows, validations and interactions.

This structured approach is conforming to best practice in system modelling and guarantees efficiency, compliance, and scalability (Sommerville, 2016).

Level 0: Context Diagram

In Figure 12, the Level 0 Context Diagram shows the PMS at a high view level as a single entity. It represents:

- The external actors (users and third-party systems) interacting with the PMS.
- How data enters and leaves the system.
- System boundaries to ensure that the core functionalities are adequately encapsulated.

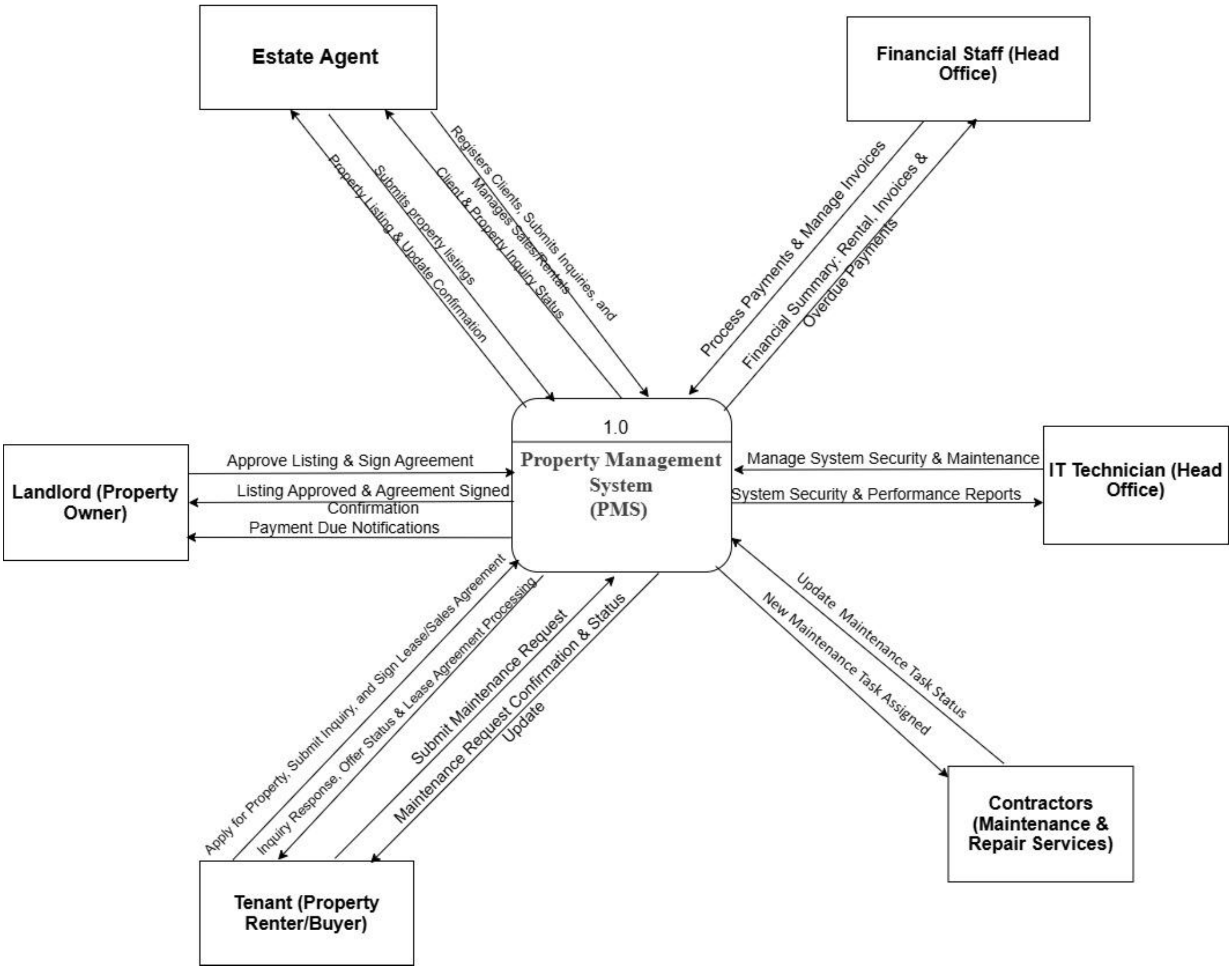


Figure 15: Level 0 Context Diagram

Key Components in the Context Diagram:

Entity	Role in the PMS	Interaction with the System
Estate Agent	Manages property listings, clients, and transactions.	Registers clients, updates listings, processes sales.
Financial Staff (Head Office)	Handles payments, invoices, and compliance.	Generates invoices, tracks transactions, ensures legal compliance.
IT Technician (Head Office)	Monitors system security, performance, and access.	Manages system updates, security logs, and database integrity.
Tenant (Property Renter/Buyer)	Searches properties, submits inquiries, pays rent.	Views listings, applies for rentals, makes payments.
Landlord (Property Owner)	Lists properties, receives rental payments.	Approves property listings, signs contracts, manages leases.
Contractors (Maintenance Services)	Provides property maintenance and repairs.	Receives maintenance requests, updates task status.

Figure 16:Level 0 Context Diagram

The diagram aids stakeholders in seeing system scope and external dependencies. It also provides a starting point for further breakdown into Level 1 and 2 DFDs (Kazi & Kazi, 2022).

Level 1: PMS Subsystems

The Level 1 DFD (Figure 13) builds on the Context Diagram by depicturing the PMS through four primary functional areas. This modular design ensures:

- Easier understanding of how different processes interact with users and databases.
- Enhanced data security and better role-based access control.
- Increased scalability for future system enhancements (Valacich et al., 2015).

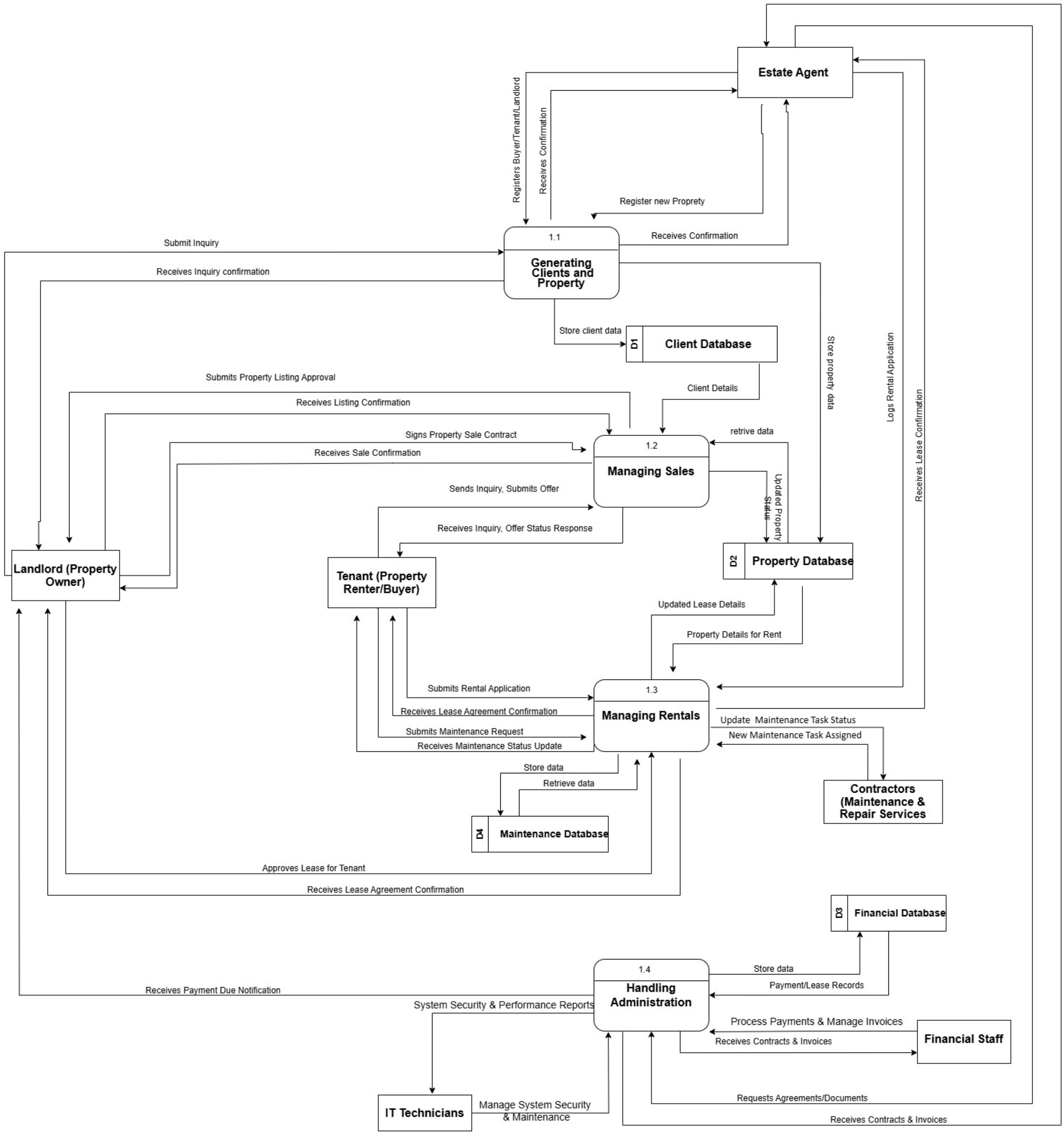


Figure 17:Level 1 DFD for PMS

PROCESS	KEY INPUTS & OUTPUTS	FUNCTION
Generating Clients and Property (1.1)	**Inputs:** Client inquiries, property data **Outputs:** Client records, property listings	Registers new clients and properties.
Managing Sales (1.2)	**Inputs:** Buyer inquiries, sales negotiations **Outputs:** Sales contracts, transaction records	Handles property sales and offers.
Managing Rentals (1.3)	**Inputs:** Tenant applications, rental payments **Outputs:** Lease agreements, maintenance records	Manages lease applications and payments.
Handling Administration (1.4)	**Inputs:** Financial data, security logs **Outputs:** Invoices, compliance reports	Manages financial operations and compliance.

Figure 18: Core Processes in Level 1 DFD

The Client Database, Property Database, Financial Database and Maintenance Database are all internal databases of each subsystem and Estate Agents, Tenants, Landlords, Contractors, and Financial Staff are external users of the above databases.

This structure is modular, scalable and helps to reduce data inconsistency across branches in line with the recommendation by GOV.UK (2018).

Level 2: Detailed Process Breakdown

The Level 2 DFDs provide a granular breakdown of the core processes to:

- To guarantee step by step precision in data flow.
- To identify validation rules, exception handling and security measures.
- To optimize automation and minimize manual errors (Kazi & Kazi, 2022).

Generating Clients and Property Listings (Level 2 DFD)

Subprocess	Inputs	Processing	Outputs
Register Clients	Client inquiry	Validate and store in **Client Database**	Confirmation message
Submit Property Listings	Property details and images	Validate and sync with external websites	Approved listing
Link Landlord with Property	Landlord approval and contract signing	Store approval details	Linked landlord-property record
Notify Tenants and Buyers	Inquiry submission	Match preferences with properties	Property match notification

Figure 19: Subprocesses and Justifications of the Generating Clients and Listings.

This process provides for the automated and structured client onboarding, thus reducing manual errors and inefficiencies. (Chakraborty, 2010).

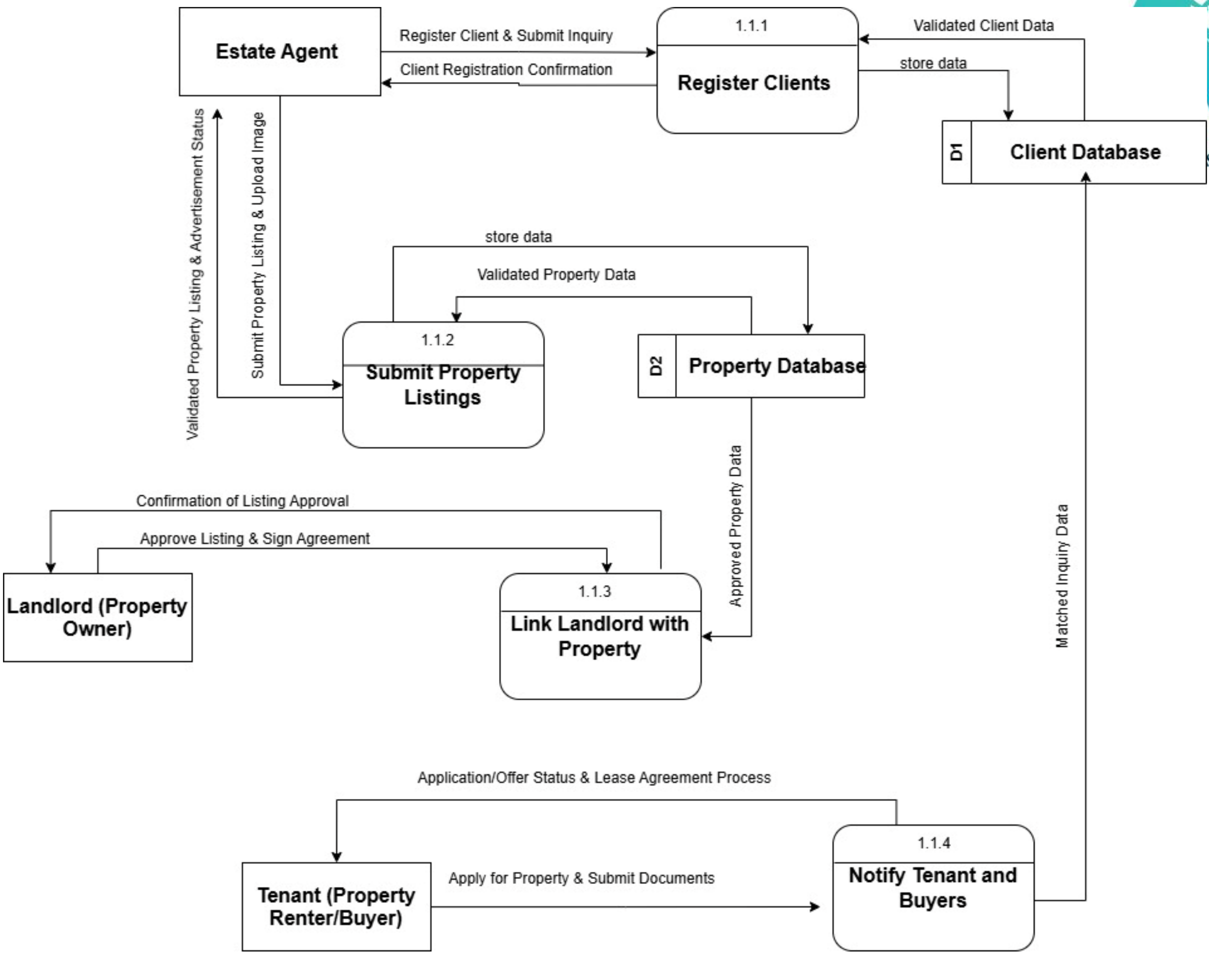


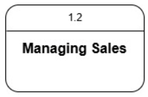
Figure 20: Level 2 DFD for "Generating Clients and Property Listings."

Managing Sales (Level 2 DFD)

Subprocess	Inputs	Processing	Outputs
Record Buyer Inquiry	Buyer details	Validate and store in **Client Database**	Inquiry confirmation
Retrieve Property Information	Buyer request	Match with property database	Matched property list
Track Offer & Negotiation	Buyer offers	Store negotiation details	Offer records
Generate Sales Contract	Accepted offer	Auto-generate contract	Sales contract for signature
Complete Sale and Update Property Status	Signed contract	Update **Property Database**	Sale completion notification

Figure 21: Level 2 DFD Subprocesses and Justifications

By using this, sales can be tracked automatically, this reduces the need for paperwork and increases efficiency (Sommerville, 2016).



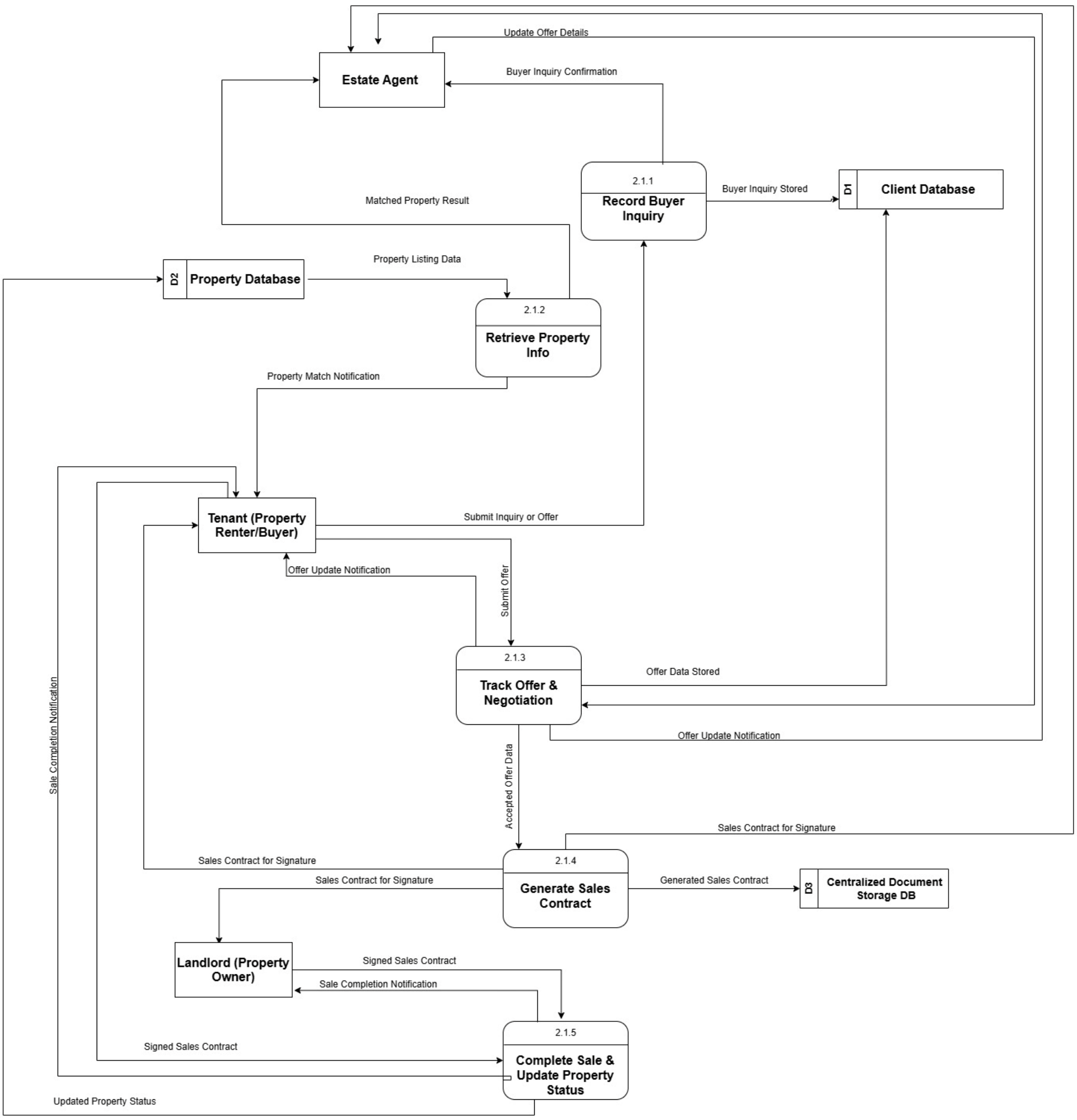


Figure 22: Level 2 DFD for Manage Sales Process.

Managing Rentals (Level 2 DFD)

Subprocess	Inputs	Processing	Outputs
Record Tenant Inquiry	Rental request	Validate and store in **Tenant Database**	Inquiry confirmation
Process Rental Application	Application documents	Generate lease agreement	Agreement for signing
Track Rental Payments	Payment details	Store in **Financial Database**	Payment receipt
Manage Maintenance Requests	Tenant request	Assign contractor	Maintenance status update

Figure 23: Subprocesses and Justifications

This increases rental tracking and property maintenance, benefiting the tenant (Valacich et al., 2015).

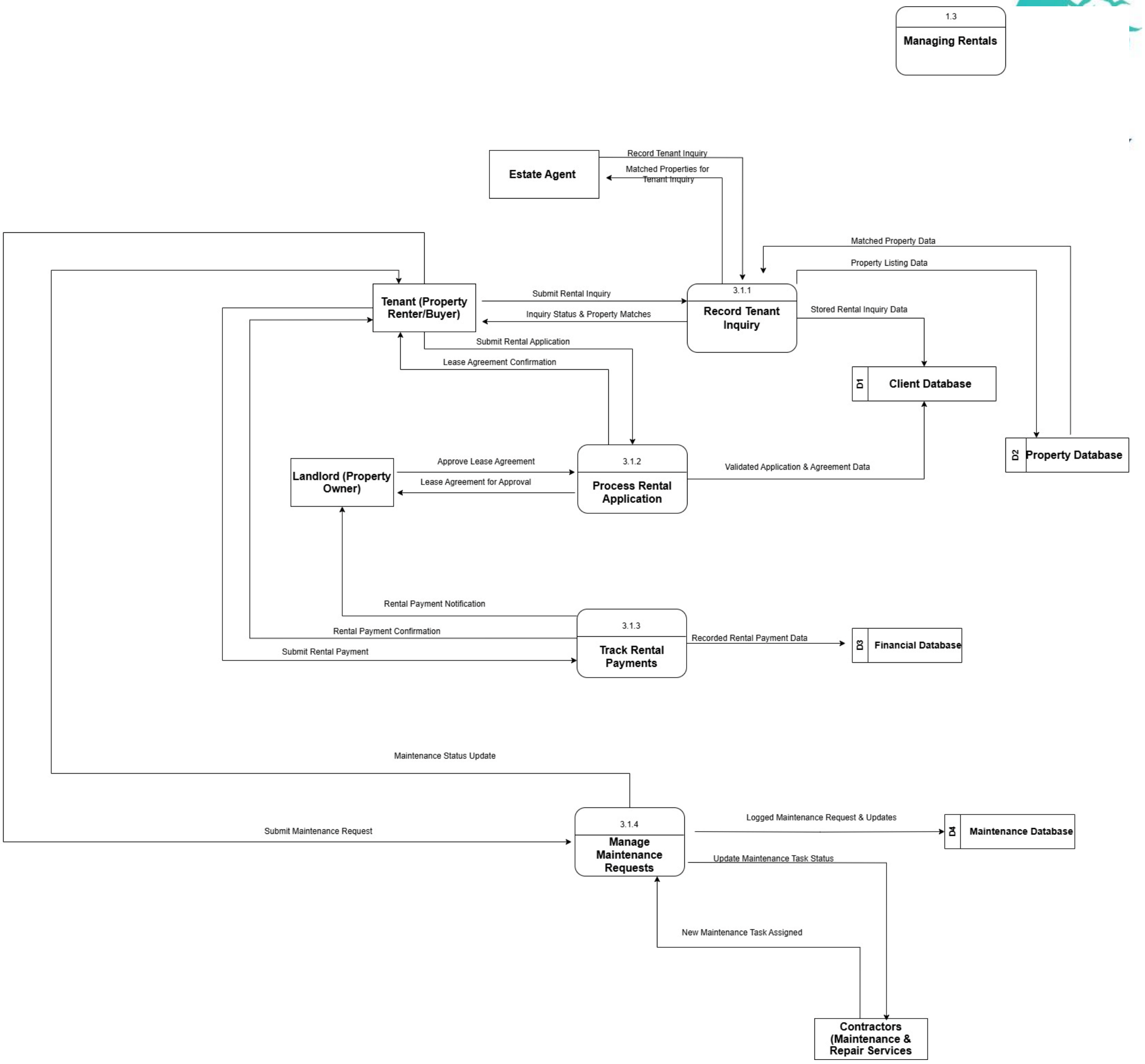


Figure 24: Level 2 DFD for Manage Rentals

Handling Administration (Level 2 DFD)

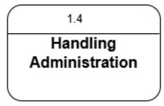
This is also a part of the PMS, which is quite helpful in:

- A financial point of view and for transaction tracking.
- Compliance in legal aspects such as GDPR and contract management.
- User authentication and cybersecurity monitoring as recommended by GOV.UK (2018).

Subprocess	Inputs	Processing	Outputs
Manage Invoices and Payments	Tenant/Landlord payments	Validate & store in **Financial Database**	Payment confirmation
Retrieve Contracts and Documents	Access request	Validate role & retrieve from **Document Storage**	Retrieved document
Ensure Data Security and GDPR Compliance	Security policy update	Audit logs & restrict access	Security status update

Figure 25: Breakdown of Subprocesses

By implementing these technologies businesses can ensure secure transactions, legal compliance, and sound financial integrity according to Sommerville (2016).



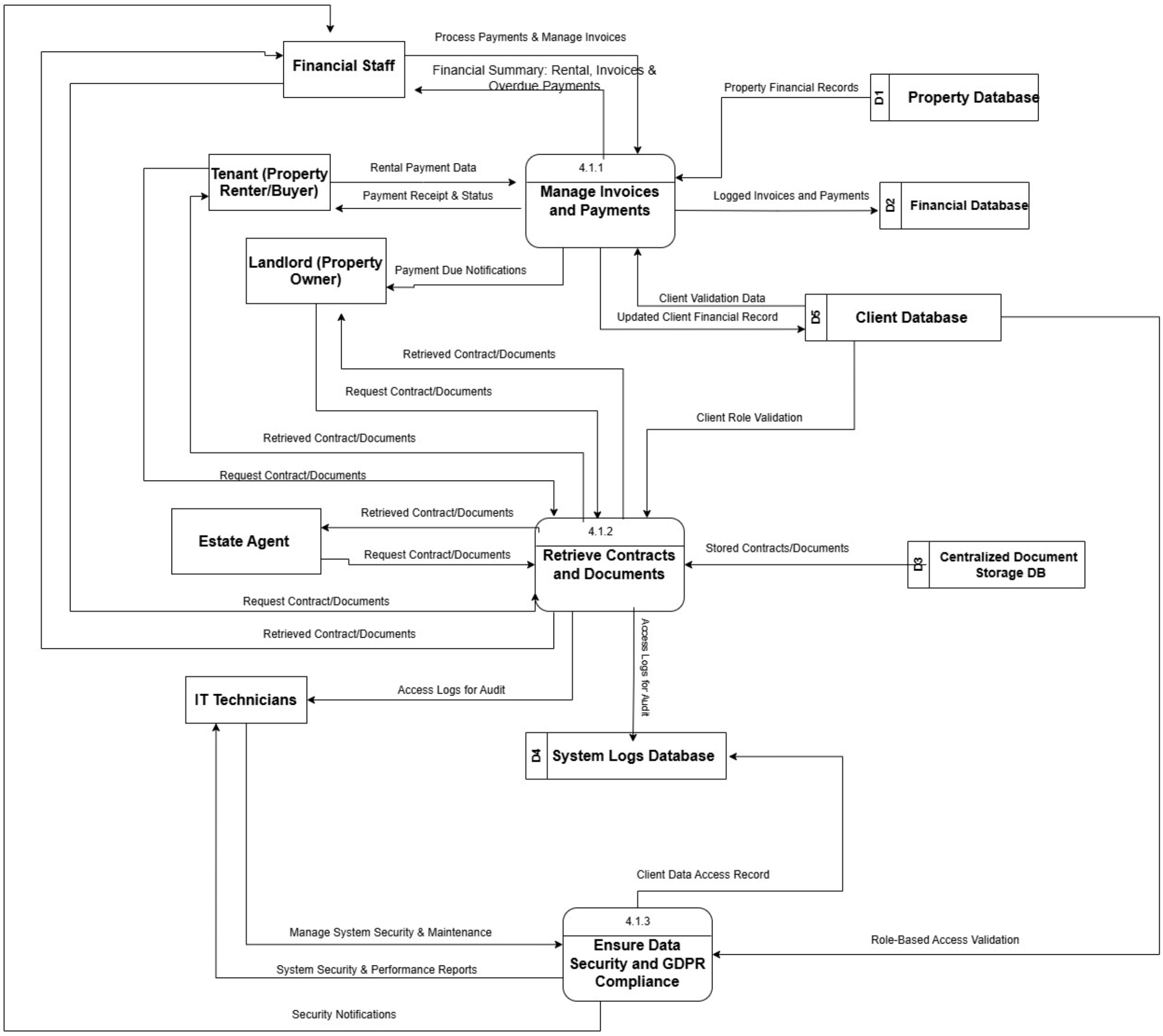


Figure 26: Level 2 DFD for Handling Administration

DFD Principles & Benefits for Prime Estates Ltd.

Industry Design Principles Applied in DFDs

Principle	Benefit
Process Clarity	All functions are well-defined and modular
Data Automation	Reduces manual errors and enhances security
Scalability & Integration	Ensures future growth and third-party system compatibility
Legal & Security Compliance	Meets GDPR standards and secures data access

The PMS benefits from the DFD:

- Clearly outlines the workflows for client management, sales, rentals, and administration.
- Ensures data storage security and compliance with financial and privacy laws.
- Developed to meet future business expansion needs.
- Structured and modular design is used to implement it, which also ensures efficiency, security, and scalability. (Kendall & Kendall 2019)

Implementation Plan for the Property Management System (PMS)

To guarantee the successful deployment and adoption of the Property Management System (PMS) at Prime Estates Ltd., a structured implementation plan is crucial. This plan follows a phased approach and includes a comprehensive testing strategy, risk management, post-deployment monitoring, stakeholder communication, and structure training to guarantee long-term operational success.

1. Implementation Strategy

To limit the potential disruptions and to permit ongoing enhancements, a phased rollout will be used to deploy the PMS (Valacich et al., 2015). The incremental approach is aligned with the Agile SDLC, which enables early delivery and user adaptation (Kendall & Kendall, 2019).

Phase	Timeframe	Activities
Phase 1 (Weeks 1-8)	Pilot deployment at one branch	Collect feedback, identify issues, optimize system functionality
Phase 2 (Weeks 9-16)	Expansion to all branches	Implement system refinements based on pilot feedback
Phase 3 (Weeks 17-24)	Full deployment	System-wide rollout, performance tuning, post-implementation monitoring

Figure 27: Deployment Phases

Key Consideration: This will enable the collection of early user engagement and iterative feedback to improve usability and functionality.

---

2. System Testing Plan

A structured testing approach is a way of ensuring that the PMS is stable, secure, and fully functional before deployment. This plan includes Unit testing, Integration testing, System testing, Security testing and User Acceptance Testing (UAT) with clearly specified testing metrics as mentioned by Ibrahim et al. (2022).

Testing Type	Objective	Metrics & Success Criteria	Timeframe
Unit Testing	Verify individual modules work correctly	≥ 95% pass rate, ≤ 500ms response time	Weeks 3-6
Integration Testing	Ensure modules interact seamlessly	API success rate ≥ 98%, data synchronization ≤ 2 sec	Weeks 7-10
System Testing	Validate full system functionality	System uptime ≥ 99.9%, UI load time ≤ 2 sec	Weeks 11-14
Security Testing	Ensure system is protected against threats	No SQL injection detected, MFA compliance	Weeks 15-18
User Acceptance Testing (UAT)	Validate usability and performance	User satisfaction score ≥ 90%, error rate ≤ 2%	Weeks 19-22

Figure 28: Testing Phases & Metrics

Automated Testing Tools: propose for use Postman for API testing, Selenium for UI testing, and AWS CloudWatch for performance monitoring.

This is because, for instance, testing can help to prevent a critical failure such as the TSB Bank IT meltdown (Peachy, 2018), which would have a dire consequence to a company like Prime Estates as far as disruptions are concerned.

---

3. Risk Management & Contingency Planning

To mitigate risks during PMS implementation, the following risk management strategies will be adopted:

Risk	Potential Impact	Mitigation Strategy
Resistance to Change	Staff may struggle to adopt the new system due to limited technical skills.	Provide structured training, step-by-step guides, and assign digital champions for support (Cadle, 2014).
Data Migration Errors	Data loss or corruption when transitioning from local databases.	Validate data, conduct pilot testing, and maintain backup copies (National Cyber Security Centre, 2019).
Integration Failures	Incorrect synchronization with property listing websites may cause listing errors.	Implement integration testing, real-time monitoring, and automated error handling (Kendall & Kendall, 2019).
Security & Compliance Risks	GDPR violations could lead to legal and financial consequences.	Use role-based access control, encrypt sensitive data, and conduct regular security audits (GOV.UK, 2018).
Project Timeline Delays	The six-month deadline may be affected by unexpected challenges.	Adopt Agile's phased implementation, prioritize core functionalities, and monitor progress iteratively (Agile Alliance, 2022).

Figure 29: Risk management strategies.

By proactively addressing these risks you can guarantee a problem free system adoption, avoid delays, and minimize disruptions.

3. Post-Implementation Support & Performance Monitoring

After its deployment, continuous monitoring and optimization will guarantee the efficiency of the PMS. It means the real-time performance tracking, automated alerts, and scalability assessments to ensure that it can deal with the increased usage (Sommerville, 2016).

Metric	Target Value	Monitoring Tool
System Uptime	≥ 99.9%	AWS CloudWatch, New Relic
API Response Time	≤ 1 sec	API Gateway logs
Error Rate	≤ 1%	Automated alerts
Page Load Time	≤ 2 sec	Lighthouse, Google PageSpeed

Figure 30: Performance Monitoring Strategy

Automated Monitoring & Alerts:

- Real-time alerts notify IT teams of performance issues.
- Monthly performance reports track stability, and real-time monitoring provides constant oversight.
- Quarterly load testing benchmarks guarantee scalability as user numbers grow.

Failure after deployment can result in financial losses and damage to a company's reputation (National Cyber Security Centre, 2019).

---

4. Stakeholder Communication Plan

The implementation process should always maintain stakeholder engagement and communication through a clear strategy as Kendall and Kendall (2019) recommend.

Stakeholder Group	Communication Method	Frequency
Estate Agents & Staff	Weekly email updates, training workshops	Weekly
IT & Development Team	Slack channel, real-time dashboards	Daily
Landlords & Tenants	Automated notifications for contracts & payments	As required
Business Owner & Executives	Monthly performance reports	Monthly

Figure 31: Stakeholder Engagement Plan

A communication plan that is well-planned will reduce confusion, get buy-in and reduce resistance to change according to Cadle (2014).

---

5. Training Plan & Adoption Strategy

As a result of the limited technical expertise of estate agents structured training is crucial for effective system adoption (El Shazly & Durand, 1991).

Training Type	Target Audience	Delivery Method	Timeframe
Workshops	Estate agents, financial staff	Hands-on training sessions	Weeks 4-6
E-learning Modules	All users	Video tutorials, manuals	Ongoing
One-on-One Support	Staff needing extra assistance	Personalized training	Weeks 10-16
Accessibility Training	Staff with disabilities	Compliance with **WCAG 2.1**	Ongoing

Figure 32:Training Phases & Methods

Post-training assessment metrics:

- User Training Success Rate: ≥ 90%
- Reduction in Support Tickets: ≥ 30%

This makes the training more effective as it improves user confidence that leads to faster adoption and reduced operational errors (Indeed Editorial Team, 2021).

This is a controlled approach of PMS implementation that ensures a successful deployment of the system, with an incremental rollout plan, a rigorous testing structure, continuous monitoring of the system, a well-planned training process and proper communication.

Using quantifiable performance metrics, automated system monitoring, stakeholder engagement, Prime Estates Ltd. is able to guarantee a smooth change over to the new system with little or no disturbance and increased productivity in the long run. (Kendall & Kendall, 2019).

Conclusion

It is imperative that Prime Estates Ltd. succeeds in implementing a centralized Property Management System (PMS) to address the inefficiencies that result from data management fragmentation, manual processing, and limited system integration (Sommerville, 2016). The choice of the Agile SDLC model provides flexibility, stakeholder involvement, and phased deployment that matches the organization’s constraints and objectives as defined by Agile Alliance (2022).

The requirements analysis was followed by the identification of core functional and non-functional needs, which included automation, CRM integration, financial tracking, and legal compliance (Valacich et al., 2015). The system design as depicted by the Data Flow Diagrams (DFDs) ensures that the architecture is structured, scalable, and secure (Kendall & Kendall, 2019).

A phased implementation plan coupled with thorough testing, stakeholder management, and training will reduce risks and improve the system's acceptance (Peachy, 2018). Moreover, performance accounting for and AI optimization of the system will help achieve long-term efficiency as well as meet GDPR and accessibility standards as per GOV.UK (2018).

This report provides a strategic, technically feasible and business aligned solution to Prime Estates Ltd., integrating current software engineering principles to support the sustainable growth and competitive advantage of the organization in the UK property market (Sommerville, 2016).

References

Agile Alliance (2022) *The Agile Manifesto*. Available at: https://www.agilealliance.org/agile101/the-agile-manifesto (Accessed: 10 February 2025).

Cadle, J. (2014) *Developing Information Systems: Practical Guidance for IT Professionals*. Swindon: BCS, The Chartered Institute for IT.

Chakraborty, P. (2010) *Design and Analysis Using Software Engineering Techniques: A Case Study*.

Fox, T. L. (2017) *Systems Analysis, Design, And Development Case Study: Sarah's Short Cakes - Sales Order System*.

GOV.UK (2018) *Data Protection Act 2018*. Available at: https://www.gov.uk

GOV.UK (2022) *Agile and government services: an introduction*. Available at: https://www.gov.uk/service-manual/agile-delivery/agile-government-services-introduction (Accessed: 10 February 2025).

Hendrix, T. D. and Schneider, M. P. (2002) 'NASA's TReK Project: A Case Study in Using the Spiral Model of Software Development', *Communications of the ACM*, 45(4), pp. 152-159. Available at: https://search.ebscohost.com (Accessed: 10 February 2025).

Ibrahim, R., Amin, A., Jamel, S., & Wahab, J. (2022) 'EPiT: A Software Testing Tool for Generation of Test Cases Automatically'.

Improving (2023) *The Seven Phases of the System-Development Life Cycle*. Available at: https://www.improvingatlanta.com/knowledge-center/system-development-life-cycle (Accessed: 10 February 2025).

Kazi, Z. and Kazi, L. (2022) 'Software Project Duration Estimation Based on COSMIC Method Applied to Data Flow Diagram'.

Kendall, K. E. and Kendall, J. E. (2019) *Systems Analysis and Design, Global Edition*. 9th edn. Pearson. Available at: https://links.arden.ac.uk/s5g72b (Accessed: 10 February 2025).

Microsoft (2022) *Security Development Lifecycle (SDL)*.

National Cyber Security Centre (2019) *Secure Design Principles*.

Peachy, K. (2018) 'TSB investigated over IT meltdown', *BBC News*, 4 August. Available at: https://www.bbc.co.uk/news/business-44370802 (Accessed: 10 February 2025).

Sommerville, I. (2016) *Software Engineering, Global Edition*. 10th edn. Pearson. Available at: https://links.arden.ac.uk/42h5xh (Accessed: 10 February 2025).

Valacich, J. S., George, J. F. and Hoffer, J. (2015) *Essentials of Systems Analysis and Design, Global Edition*. 6th edn. Pearson. Available at: https://links.arden.ac.uk/rxxt7g (Accessed: 10 February 2025).